# 1. Introduction

The rapid advancement of digital technologies and artificial intelligence has significantly transformed the way individuals interact with information and make decisions. Modern digital environments such as social media platforms, recommendation systems, online marketplaces, and analytics dashboards require users to process large volumes of information and make frequent decisions in short periods of time. While these technologies offer convenience and efficiency, they also increase the cognitive demands placed on users. As a result, understanding how digital environments influence human cognition and decision-making has become an important area of research in fields such as psychology, human–computer interaction (HCI), and data science.

One of the key concepts used to understand this phenomenon is **cognitive load**, which refers to the amount of mental effort required to process information and perform tasks. When individuals interact with complex digital interfaces or are exposed to excessive information, their cognitive resources may become overloaded. This can negatively affect attention, information processing ability, and decision-making performance. In digital systems where users are continuously required to evaluate options, respond to notifications, and interpret data, the accumulation of cognitive load can lead to mental exhaustion.

Closely related to cognitive load is the concept of **decision fatigue**. Decision fatigue occurs when the quality of decisions deteriorates after prolonged decision-making activity. Research in behavioural psychology suggests that human decision-making capacity is limited, and repeated decisions gradually deplete mental energy. As this mental energy decreases, individuals are more likely to avoid decisions, rely on default choices, or make impulsive judgments. In digital environments where users are required to make numerous micro-decisions—such as selecting content, responding to messages, or analysing information—decision fatigue can occur more rapidly.

Another important factor influencing digital decision-making is the increasing use of **algorithmic systems and artificial intelligence**. Many digital platforms use recommendation algorithms and predictive models to assist users in making decisions. While these systems aim to reduce cognitive effort by providing suggestions or automated insights, research shows that users do not always trust algorithmic recommendations. In some cases, users may reject algorithmic advice after observing errors, a phenomenon commonly referred to as **algorithm aversion**. This interaction between human cognition and automated systems adds another layer of complexity to decision-making in digital environments.

Several researchers have examined how prolonged digital interaction, information overload, and algorithmic assistance affect cognitive load and decision-making behaviour. Studies have explored topics such as Zoom fatigue, algorithm aversion, decision fatigue, and cognitive workload in digital dashboards. These studies highlight the growing concern that continuous digital engagement may negatively influence users' mental wellbeing and decision quality.

Despite the growing body of research in this area, many existing studies focus primarily on conceptual explanations or controlled experimental settings rather than analysing real behavioural data from digital platforms. In addition, limited research has applied computational methods or machine learning techniques to quantitatively measure decision fatigue based on user interaction patterns. Understanding how behavioural signals such as interaction timing, navigation behaviour, and information processing patterns relate to cognitive load could provide valuable insights for designing more user-friendly digital systems.

Therefore, this literature review aims to examine previous research related to cognitive load, decision fatigue, and algorithmic decision-making in digital environments. The review analyses key findings from existing studies, compares their methodologies and limitations, and identifies research gaps that motivate further investigation. By synthesising insights from psychology, human computer interaction, and data science, this review provides a foundation for understanding how digital systems influence human decision behaviour and highlights opportunities for future research in this area.

# 2. Background Concepts

## 2.1 Cognitive Load Theory

Cognitive Load Theory explains how individuals process information and perform tasks using their working memory. According to this theory, the human brain has a limited capacity for processing information at any given time. When the amount of information exceeds this capacity, cognitive overload occurs, which reduces an individual's ability to effectively process information and make accurate decisions.

Cognitive load can be categorized into three main types: intrinsic cognitive load, extraneous cognitive load, and germane cognitive load. Intrinsic cognitive load refers to the inherent difficulty of a task or information being processed. Extraneous cognitive load is generated by the way information is presented to the user, such as poorly designed interfaces or complex visual layouts. Germane

cognitive load relates to the mental effort used to understand and learn new information.

In digital environments, cognitive load is often influenced by the design of user interfaces, the complexity of information presented, and the number of tasks users are required to perform simultaneously. For example, complex dashboards, multiple notifications, and large volumes of data can significantly increase cognitive load. When users experience high cognitive load, their ability to concentrate, evaluate options, and make well-informed decisions decreases.

Understanding cognitive load is particularly important in the context of digital platforms, as poorly designed systems may overwhelm users with excessive information. Researchers in human–computer interaction emphasize that reducing unnecessary cognitive load through better interface design can improve user performance, reduce mental fatigue, and enhance decision-making efficiency.

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## 2.2 Decision Fatigue

Decision fatigue refers to the decline in the quality of decisions made by individuals after a prolonged period of decision-making. Psychological research suggests that human decision-making relies on a limited amount of mental energy. As individuals continue making decisions over time, this mental energy gradually becomes depleted, resulting in reduced decision-making effectiveness.

When individuals experience decision fatigue, they tend to rely on simplified decision strategies. This may include avoiding decisions altogether, selecting default options, or making impulsive choices without carefully evaluating available alternatives. Decision fatigue has been observed in various contexts, including consumer behaviour, financial decision-making, and digital interactions.

In digital environments, users are often required to make numerous small decisions throughout their interaction with a system. For instance, users may need to select content, evaluate recommendations, respond to notifications, and interpret data visualizations. Although each individual decision may appear minor, the accumulation of many such decisions can significantly increase mental workload and lead to decision fatigue.

Studies suggest that decision fatigue not only affects individual performance but can also influence productivity and overall user experience. As digital platforms

continue to expand and offer more interactive features, understanding the impact of decision fatigue has become increasingly important for designing systems that support efficient and sustainable decision-making.

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## 2.3 Algorithm Aversion

Algorithm aversion refers to the tendency of individuals to distrust or avoid algorithmic recommendations after observing errors in automated systems. Although algorithms are often capable of making accurate predictions and recommendations, users may lose confidence in them if they perceive them as unreliable or difficult to understand.

Research shows that individuals often hold algorithms to higher standards than human decision-makers. While people may tolerate occasional mistakes made by humans, they tend to react more negatively when errors are made by algorithms. As a result, users may prefer to rely on their own judgment rather than follow algorithmic recommendations, even when algorithms perform better on average.

Algorithm aversion is particularly relevant in digital platforms where artificial intelligence and machine learning systems are used to guide user decisions. Recommendation systems in e-commerce, content platforms, and data analytics tools frequently provide suggestions intended to simplify decision-making. However, if users do not trust these systems, they may ignore recommendations, reducing the effectiveness of automated decision-support technologies.

Understanding algorithm aversion is important for improving the design of AI-driven systems. Researchers emphasize the importance of transparency, explainability, and user control in increasing trust in algorithmic systems and encouraging users to adopt automated recommendations.

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## 2.4 Digital Decision Environments

Digital decision environments refer to platforms where users interact with digital interfaces to evaluate information and make choices. These environments include online shopping platforms, social media applications, recommendation systems, and data analytics dashboards.

In such environments, users are often exposed to large volumes of information that must be processed quickly. Features such as personalized

recommendations, interactive visualizations, and constant notifications can increase the number of decisions users must make during a single session. While these features are designed to improve usability and engagement, they may also contribute to cognitive overload and mental fatigue.

Human–Computer Interaction research highlights that the design of digital environments plays a crucial role in shaping user behaviour. Factors such as interface complexity, information density, navigation structure, and interaction patterns can influence how users process information and make decisions.

As digital systems continue to evolve, researchers are increasingly interested in understanding how user interaction data can be used to measure cognitive load and decision fatigue. By analysing behavioural signals such as click patterns, response times, and interaction sequences, it may be possible to identify when users experience mental fatigue and design systems that adapt accordingly.

# 3. Review of Previous Studies

Several researchers have studied the relationship between digital interaction, cognitive load, and decision-making behaviour. With the rapid growth of digital platforms, individuals are increasingly required to process large amounts of information and make frequent decisions. Researchers have examined how prolonged digital interaction, algorithmic systems, and complex interfaces affect cognitive load and decision fatigue. The following studies provide important insights into how digital environments influence mental workload and decision-making patterns.

| Title | Author(s) | Year | Key Findings | Limitations |
|---|---|---|---|---|
| Zoom Fatigue: A Theoretical Analysis | Jeremy N. Bailenson | 2022 | Long video conferencing sessions increase cognitive load and mental fatigue. Reduced non-verbal cues and constant screen interaction lead to mental strain and decreased decision-making ability. | The study mainly focuses on video conferencing platforms rather than general digital environments. |
| Algorithm Aversion Revisited: People Avoid Algorithms After Errors | Berkeley J. Dietvorst, Joseph P. Simmons, Cade Massey | 2023–2024 | The study shows that users change their decision behaviour when interacting with algorithms. Trust and mental fatigue influence reliance on AI recommendations. | The research does not incorporate temporal interaction modelling or real behavioural datasets. |

| Title | Author(s) | Year | Key Findings | Limitations |
|---|---|---|---|---|
| Decision Fatigue in the Digital Era: Are AI Solutions Reducing Cognitive Load? | Kathleen D. Vohs, Roy F. Baumeister | 2025 | Continuous digital decision- making reduces self-control and may negatively affect productivity. AI recommendations may either reduce or increase decision fatigue depending on system design. | The study is primarily conceptual and does not use clickstream or behavioural data. |
| Cognitive Load and User Engagement in Real-Time Analytics Dashboards | Various HCI & Data Science Researchers | 2026 | Complex dashboards and information-heavy interfaces increase mental workload.<br>Information overload leads to poor decision quality and reduced user engagement. | Real-time predictive models for decision fatigue are still limited. |
| Avoiding Decision Fatigue with AI-Assisted Decision-Making | Jessica Maria Echtehof, Aditya Melkote, Sujan Kanchela, Julian McAuley | 2023 | The study shows that AI-assisted systems reduce cognitive load and improve decision efficiency. It highlights that intelligent recommendations help minimize mental fatigue in digital environments. | The study may reduce user control due to over-reliance on AI. It does not consider temporal interaction patterns for fatigue modelling. |
| Combating Decision Fatigue and Optimising Decision-Making in Information Technology (IT): The Role of Artificial Intelligence (AI) in Enhancing Cognitive Outcomes | Palitha Jayasanka Perera Angage | 2025 | The study shows that decision fatigue affects performance and AI tools help improve cognitive outcomes.<br>It highlights that intelligent systems optimize decision-making in complex IT environments. | The study is limited to IT-specific applications and lacks generalization to digital platforms. It does not use behavioural or temporal data for modelling decision fatigue |
| Predict Fatigue Using Smartphone Usage Data | Yaqi Yang | 2024 | The study shows that smartphone usage patterns can predict fatigue using temporal features. It highlights that machine learning models effectively analyze behavioural interaction data. | The study focuses on general fatigue rather than decision fatigue. It does not directly apply findings to digital platform decision-making scenarios. |

## Discussion of Previous Studies

The reviewed studies highlight the growing concern regarding cognitive load and decision fatigue in digital environments. Bailenson (2022) examined the phenomenon of Zoom fatigue and explained how prolonged video conferencing sessions increase mental strain due to constant screen interaction and reduced non-verbal communication cues. The study emphasized that digital interface design can significantly influence attention and decision-making performance.

Dietvorst, Simmons, and Massey (2023–2024) investigated how individuals respond to algorithmic recommendations. Their findings indicate that users may avoid relying on algorithms after observing errors, even when algorithms provide useful decision support. This behaviour reflects the complex relationship between trust, human judgement, and automated systems.

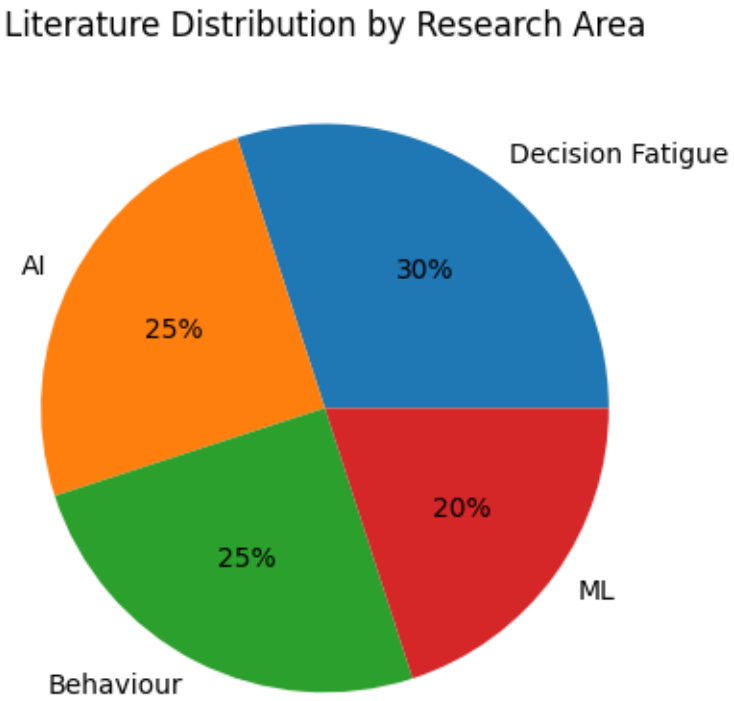
Research by Vohs and Baumeister (2025) focused on decision fatigue in modern digital environments. Their work suggests that repeated decision-making tasks gradually reduce self-control and decision quality. As digital platforms require users to make numerous small decisions, the risk of decision fatigue becomes more significant.

| A | B | C | D | E | F |
|---|---|---|---|---|---|
| Paper Title | Type | Data Used | ML Used | Focus Area | Gap Identified |
| Zoom Fatigue: A Theoretical Analysis | Cognitive | No | No | Decision Fatigue | No ML, no real user data |
| Algorithm Aversion: People Avoid Algorithms After Errors | AI | No | No | Decision Behaviour | No behavioural data, no fatigue focus |
| Decision Fatigue in the Digital Era | Cognitive | No | No | Decision Fatigue | No temporal data, no ML modelling |
| Cognitive Load and User Engagement in Real-Time Analytics Dashboards | Behaviour | Yes | No | Cognitive Load | No decision fatigue modelling |
| Avoiding Decision Fatigue with AI-Assisted Decision-Making | AI | No | No | Decision Support | No behavioural data, no temporal analysis |
| Combating Decision Fatigue in IT using AI | AI | No | No | Decision Optimization | Limited to IT, no interaction data |
| Predict Fatigue Using Smartphone Usage Data | ML | Yes | Yes | Temporal Behaviour | Focuses on general fatigue, not decision fatigue |

## Comparative Analysis of Existing Studies

Recent Human–Computer Interaction research (2026) has also explored how complex digital dashboards influence user cognition. These studies demonstrate that information-heavy interfaces increase mental workload and reduce decision-making efficiency. Interaction patterns such as slower responses and increased hesitation may indicate cognitive stress during digital interaction.

Overall, the reviewed studies demonstrate that digital environments significantly influence cognitive load and decision-making behaviour. However, many of these studies focus on conceptual explanations or specific digital platforms rather than analysing real user interaction data across broader digital systems.




Distribution of Literature Based on Research Areas

# 4. Comparative Analysis of Studies

The reviewed studies collectively provide important insights into how digital environments influence cognitive load, decision fatigue, and user behaviour. Although each study focuses on a different aspect of digital interaction, several common themes emerge when comparing their findings.

One of the key similarities across these studies is the recognition that **continuous digital interaction increases cognitive demands on users**. Bailenson (2022) demonstrated that prolonged video conferencing sessions require individuals to process multiple visual and social cues simultaneously, leading to increased mental fatigue. Similarly, research on digital dashboards (2026) indicates that complex visual interfaces and excessive information presentation increase cognitive workload, which can negatively affect user performance and decision quality.

Another important theme observed across the studies is the role of **decision-making frequency in contributing to mental fatigue**. Vohs and Baumeister (2025) highlighted that repeated decision-making tasks gradually deplete mental resources, reducing self-control and decision accuracy. This finding aligns with the challenges faced by users in digital environments where they must constantly evaluate information, choose between alternatives, and respond to notifications.

The studies also highlight the impact of **algorithmic systems on user decision behaviour**. Dietvorst, Simmons, and Massey (2023–2024) demonstrated that users may lose trust in algorithms after observing even small errors. This phenomenon, known as algorithm aversion, suggests that although algorithms are designed to support decision-making, users may hesitate to rely on them when trust is reduced. As digital platforms increasingly incorporate artificial intelligence and recommendation systems, understanding user trust in algorithms becomes essential for improving system effectiveness.

Despite these similarities, the studies also differ in their **research focus and methodological approaches**. Bailenson's research primarily examines digital communication platforms such as video conferencing systems, while the dashboard studies focus on information visualization and analytics tools. In contrast, the work of Vohs and Baumeister emphasizes psychological aspects of decision fatigue, and the research by Dietvorst and colleagues focuses on behavioural responses to algorithmic decision support.

Another key difference lies in the **type of data and analysis methods used**. Many of the reviewed studies rely on conceptual frameworks or controlled experiments rather than analysing real behavioural data from digital platforms. As a result, while these studies provide valuable theoretical insights into cognitive load and decision fatigue, they do not fully capture the complexity of real-world digital interactions.

Overall, the comparative analysis suggests that cognitive load and decision fatigue are influenced by multiple factors, including interface complexity, frequency of decision-making, and trust in automated systems. Although existing research provides important theoretical foundations, further studies are required to analyse real user interaction data and develop predictive models capable of identifying decision fatigue in digital environments.

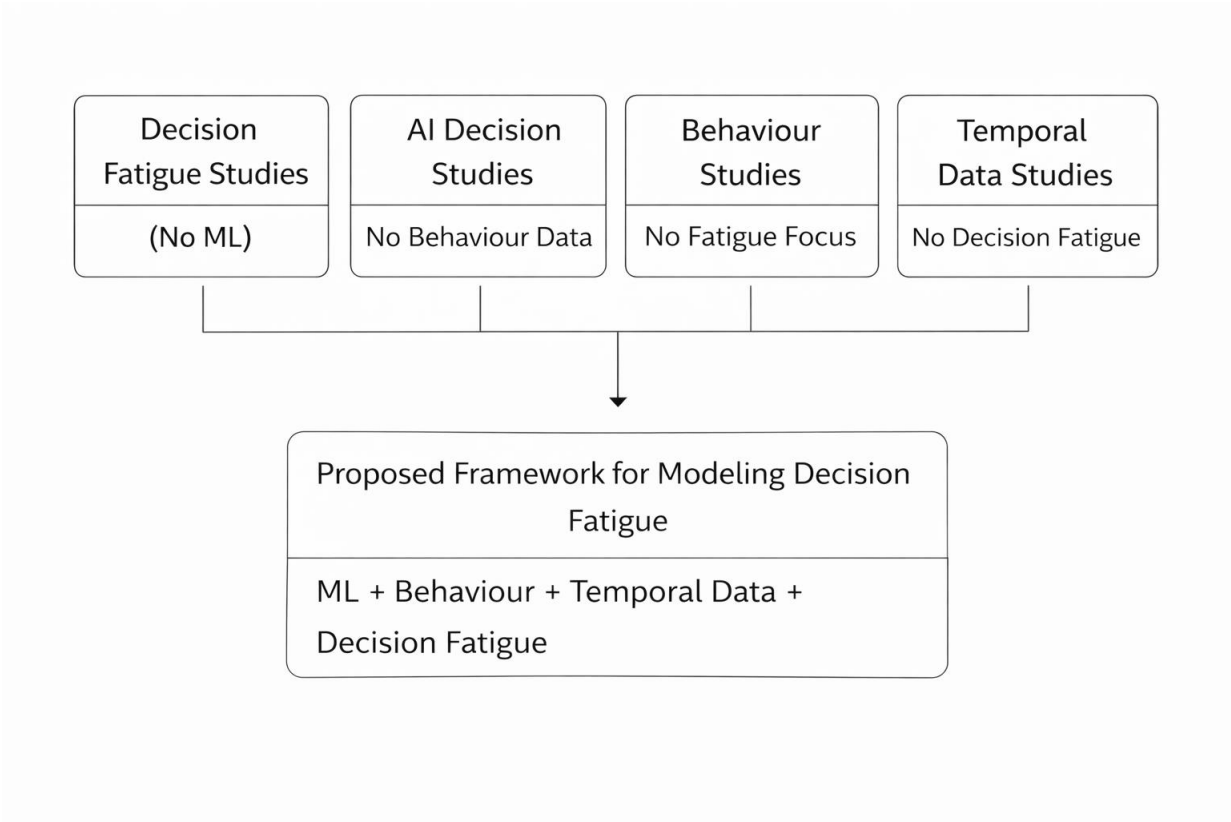
# 5. Research Gaps

Although existing research provides valuable insights into cognitive load, decision fatigue, and algorithmic interaction in digital environments, several important research gaps remain. Identifying these gaps is essential for understanding the limitations of current studies and highlighting areas where further research is required.

One of the primary gaps in the existing literature is the limited focus on real behavioural data from digital platforms. For example, Bailenson's study (2022) explains the concept of Zoom fatigue and how prolonged video conferencing increases cognitive load. While this research provides important theoretical insights, it mainly focuses on video conferencing environments and does not examine broader digital platforms where users interact with multiple systems simultaneously.

Similarly, the research on algorithm aversion by Dietvorst, Simmons, and Massey (2023–2024) highlights how users may avoid algorithmic recommendations after observing errors. Although this study provides valuable understanding of user trust in automated systems, it does not incorporate temporal interaction modelling or real-world behavioural datasets that capture how users interact with algorithms over time.

Another limitation is observed in the work of Vohs and Baumeister (2025), which discusses decision fatigue in the digital era. Their research explains how repeated decision-making reduces self-control and negatively affects




## Research Gap Identification in Existing Studies

productivity. However, the study remains largely conceptual and does not analyse behavioural interaction data, such as click patterns or decision timing, which could provide measurable indicators of decision fatigue.

Furthermore, research on cognitive load in real-time analytics dashboards (2026) demonstrates that complex interfaces and excessive information can increase mental workload. Although these studies highlight the impact of interface complexity on user cognition, they do not provide predictive models capable of identifying decision fatigue in real-time digital environments.

Another important research gap is the lack of integration between behavioural psychology and computational methods. Most existing studies examine cognitive load and decision fatigue from a theoretical or experimental perspective, but few studies attempt to apply data science techniques or machine learning models to quantitatively measure these phenomena.

In addition, many studies focus on specific digital platforms or isolated interaction scenarios rather than analysing the broader digital ecosystem where users frequently switch between multiple platforms. As modern users interact with various digital tools such as communication platforms, recommendation systems, and analytics dashboards, understanding decision fatigue across these environments remains an important challenge.

Therefore, there is a need for research that combines insights from psychology, human–computer interaction, and data science to better understand cognitive

load and decision fatigue in digital systems. Future studies should explore how real user interaction data can be used to detect behavioural patterns associated with mental fatigue and develop computational models capable of predicting decision fatigue in digital environments.

# 6. Methodology

The selected studies employ a diverse range of methodologies, including theoretical, experimental, and data-driven approaches, to investigate decision fatigue, cognitive load, and user behaviour in digital environments.

The study on *Zoom Fatigue: A Theoretical Analysis* adopts a conceptual methodology grounded in psychological theory to explain the cognitive mechanisms underlying fatigue in virtual interactions. Similarly, the research on *Decision Fatigue in the Digital Era* follows a review-based approach, synthesizing existing literature to understand the impact of prolonged decision-making in digital contexts.

The paper on *Algorithm Aversion: People Avoid Algorithms After Errors* utilizes an experimental methodology, where controlled experiments are conducted to observe user behaviour and decision-making patterns when interacting with algorithmic systems. This approach enables the analysis of trust, error perception, and decision responses.

In contrast, the study on *Cognitive Load and User Engagement in Real-Time Analytics Dashboards* employs a behavioural and observational methodology, analyzing user interaction data to assess how interface complexity influences cognitive load and engagement. This provides insights into real-world user behaviour but lacks predictive modelling.

The research on *Avoiding Decision Fatigue with AI-Assisted Decision-Making* and *Combating Decision Fatigue in IT using AI* adopt application-oriented and system evaluation methodologies, focusing on the role of artificial intelligence in enhancing decision-making efficiency and reducing cognitive burden in complex environments.

Finally, the study on *Predict Fatigue Using Smartphone Usage Data* follows a data-driven methodology, leveraging real-world smartphone usage data and applying machine learning techniques to predict fatigue levels based on temporal interaction patterns. This approach highlights the potential of integrating behavioural data with predictive modelling.

Overall, while the reviewed studies provide valuable insights into decision fatigue, cognitive load, and user behaviour, they primarily operate in isolated domains, with limited integration of temporal interaction data and machine learning techniques for comprehensive decision fatigue modelling.

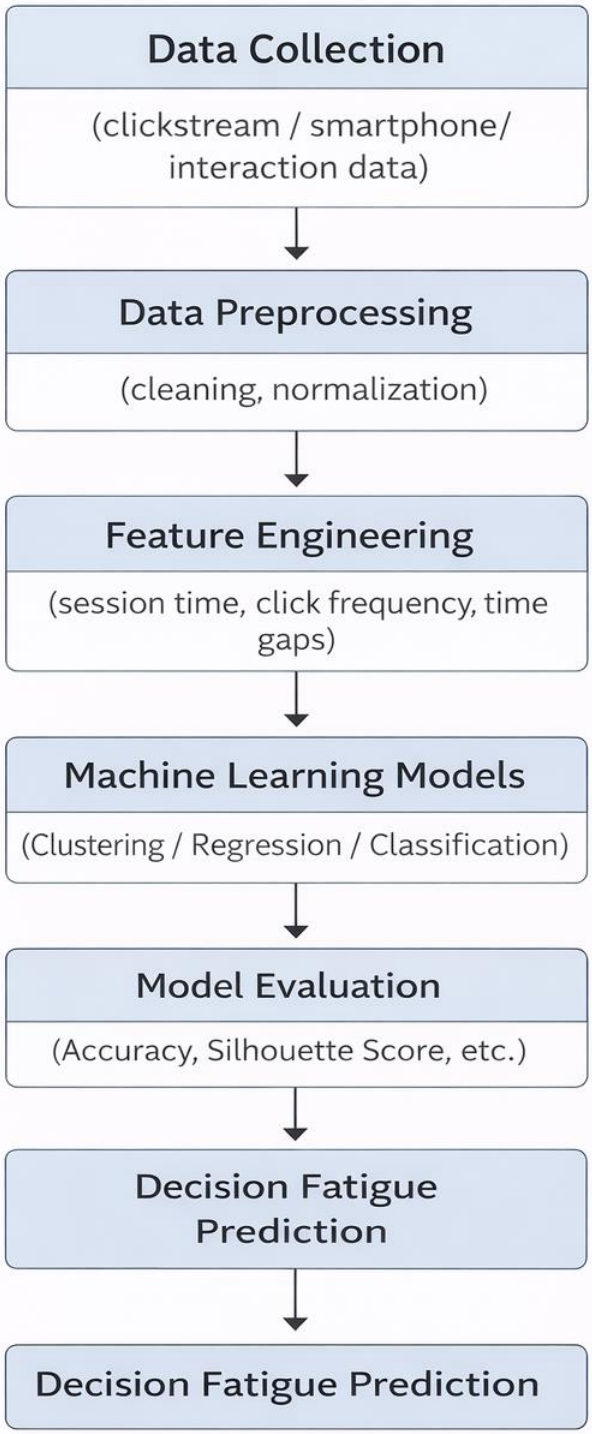



**Figure:** Proposed Methodology for Modeling Decision Fatigue Using Temporal Interaction Patterns

# 7. Conclusion

This literature review examined existing research related to cognitive load, decision fatigue, algorithm interaction, and digital decision environments. With the increasing use of digital technologies, individuals are required to process large volumes of information and make frequent decisions while interacting with digital platforms. As a result, understanding how digital systems influence human cognition and decision-making behaviour has become an important research area in fields such as psychology, human–computer interaction, and data science.

The reviewed studies highlight several important insights regarding the impact of digital environments on mental workload. Bailenson (2022) demonstrated that prolonged video conferencing can lead to cognitive overload and mental exhaustion, commonly referred to as Zoom fatigue. The research emphasizes how constant screen interaction and reduced non-verbal cues can increase mental strain and affect decision-making ability.

Similarly, the work of Dietvorst, Simmons, and Massey (2023–2024) explored how individuals interact with algorithmic decision-support systems. Their findings indicate that users may develop algorithm aversion after observing errors in automated systems, even when these systems provide useful recommendations. This highlights the importance of trust and transparency in algorithm-based decision environments.

Research by Vohs and Baumeister (2025) further explains the concept of decision fatigue, suggesting that continuous decision-making gradually depletes mental resources and reduces decision quality. In digital environments where users frequently evaluate options and make repeated choices, the effects of decision fatigue may become more pronounced.

In addition, studies related to cognitive load in digital dashboards (2026) demonstrate that complex interfaces and information overload can increase mental workload and negatively affect user engagement and decision accuracy. These findings highlight the importance of effective interface design in reducing unnecessary cognitive effort for users.

Although these studies provide valuable theoretical insights, several limitations remain. Many existing studies focus on conceptual explanations or specific digital platforms rather than analysing real behavioural data from digital systems. Furthermore, limited research has applied computational methods or predictive models to measure cognitive load and decision fatigue using user interaction data.

Overall, the literature suggests that digital environments can significantly influence cognitive load and decision-making behaviour. However, further research is needed to develop data-driven approaches that analyse real user interaction patterns and identify indicators of decision fatigue in digital systems. Such research can contribute to the development of more user-friendly digital platforms that reduce cognitive overload and support better decision-making.